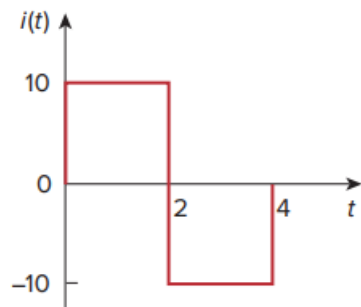


Problem

Express the current pulse in Fig. 7.33 in terms of the unit step. Find its integral and sketch it.

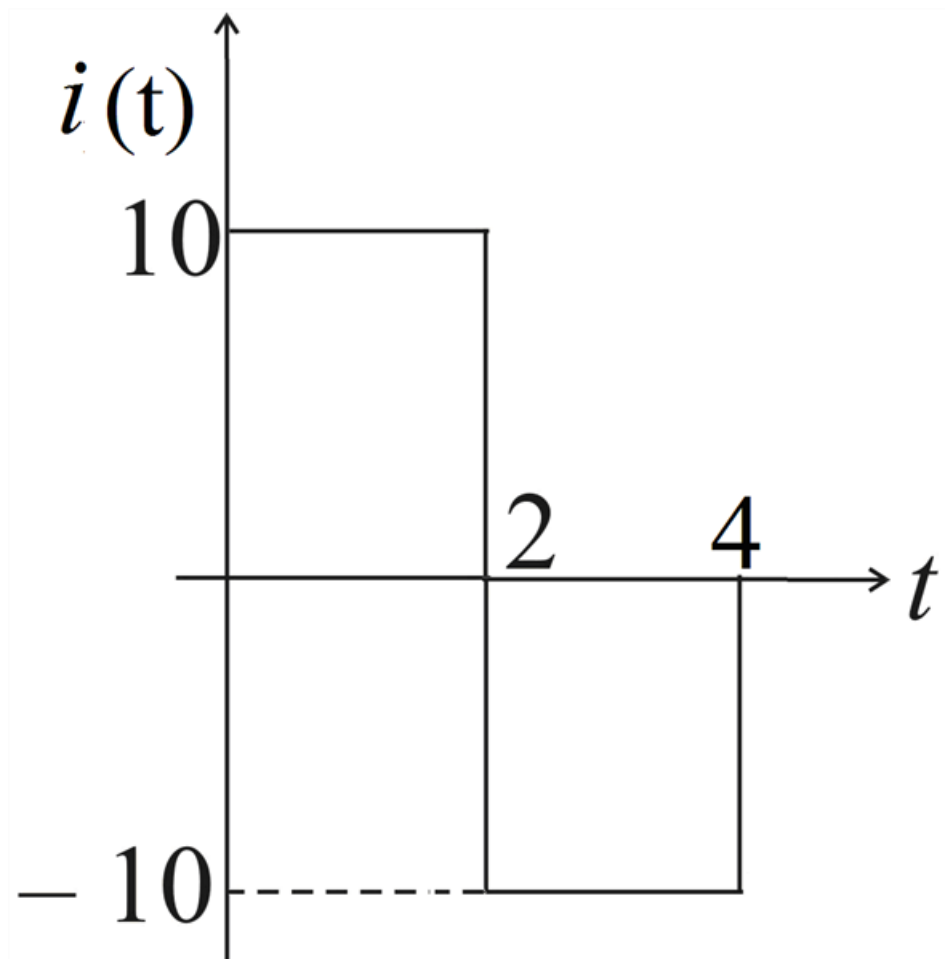
Fig. 7.33:



Step-by-step solution

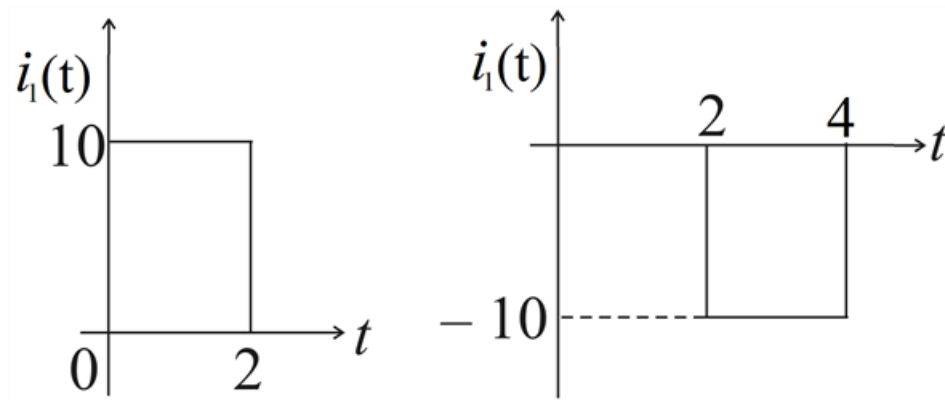
Step 1 of 6

Given current pulse



Step 2 of 6

Given pulse can be divided into two parts



Step 3 of 6

$i_1(t)$ is a step function that switches on at one value of t and switches off at another value of t .

$i_1(t)$ can be expressed as

$$i_1(t) = 10[u(t) - u(t-2)]$$

In the e way $i_2(t)$ can be expressed as

$$i_2(t) = -10[u(t-2) - u(t-4)]$$

$$\therefore i(t) = i_1(t) + i_2(t)$$

$$i(t) = 10[u(t) - u(t-2)] - 10[u(t-2) - u(t-4)]$$

$$\boxed{i(t) = 10[u(t) - 2u(t-2) + u(t-4)]}$$

Step 4 of 6

We know

$$\int_{-\infty}^t u(t) = tu(t) = r(t)$$

$$\therefore \int_{-\infty}^t u(t) = r(t)$$

By integrating $i(t)$ expression we can get

$$\int i(t) = 10 \left[\int_{-\infty}^t u(t) - 2 \int_{-\infty}^t u(t-2) + \int_{-\infty}^t u(t-4) \right]$$

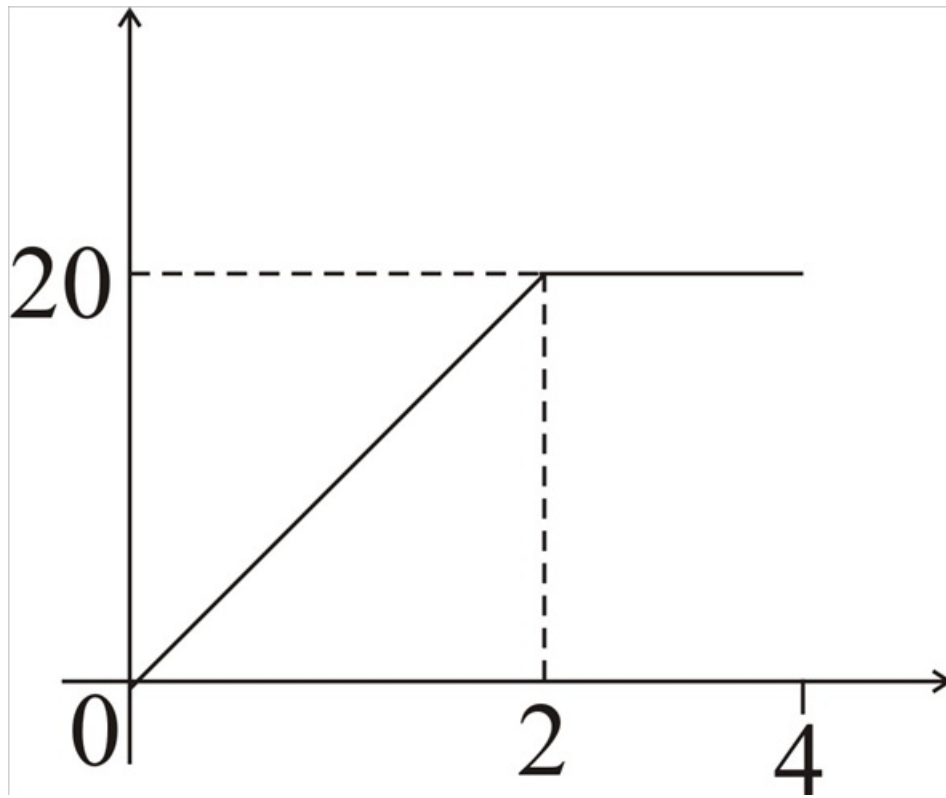
$$\boxed{\int i(t) = 10[r(t) - 2r(t-2) + r(t-4)]}$$

Above function can be represented as

$$\int i(t) = 10[r(t) - r(t-2)] - 10[r(t-2) - r(t-4)]$$

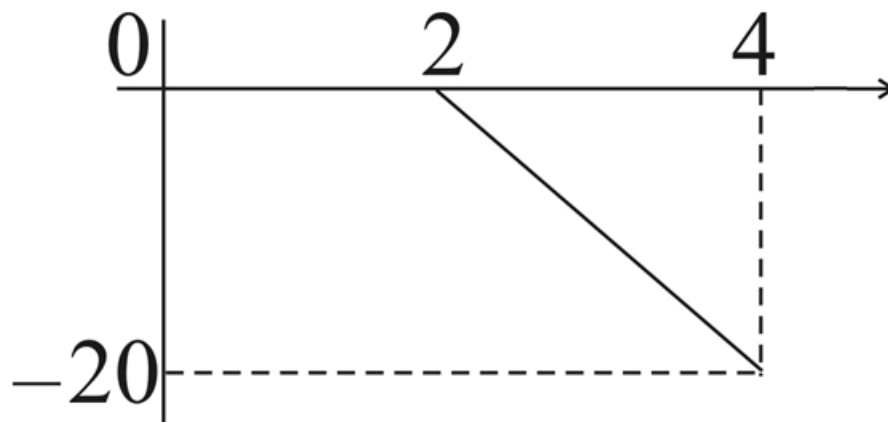
Step 5 of 6

Plot of $10[r(t) - r(t-2)]$ is



Step 6 of 6

And plot of $-10[r(t-2)-r(t-4)]$ is



$\therefore$ By combining above two plots we can get plot for two plots $\int i(t)$

